# MADHVAN E-COMMERCE SALES ANALYTICS

## Data Analytics Project Report

This project analyzes e-commerce sales data to identify trends in sales, profit, quantity, categories, regions and payment modes. Power BI is used for interactive visualization and Python is used for reproducible analysis.

**Tools:** Power BI, Excel/CSV, Python, Pandas, NumPy, Matplotlib, Scikit-learn

**Data:** Orders.csv + Details.csv | **Period:** January–December 2018

## 1. Objectives

- Measure total sales, profit and quantity.
- Identify high-performing states and categories.
- Study monthly sales and profit trends.
- Analyze payment-mode behavior.
- Create an interactive Power BI dashboard.
- Demonstrate a basic ML customer-segmentation workflow.

## 2. Dataset

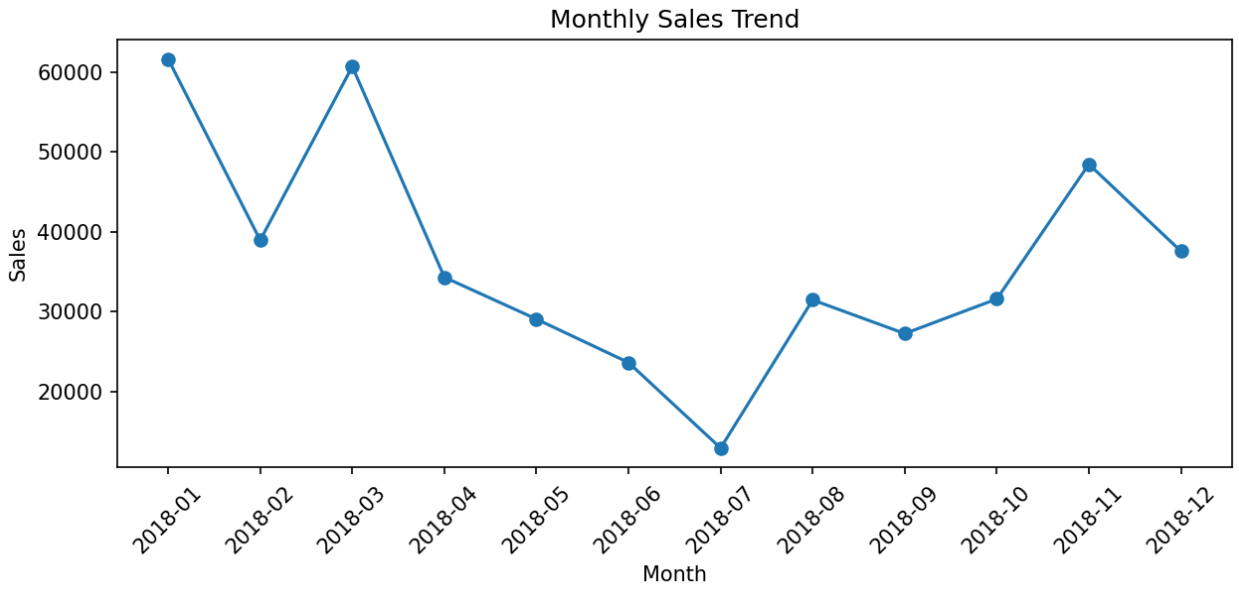
Orders.csv contains 500 orders with date, customer, state and city. Details.csv contains 1,500 detail records with amount, profit, quantity, category, sub-category and payment mode. The datasets are joined using Order ID.

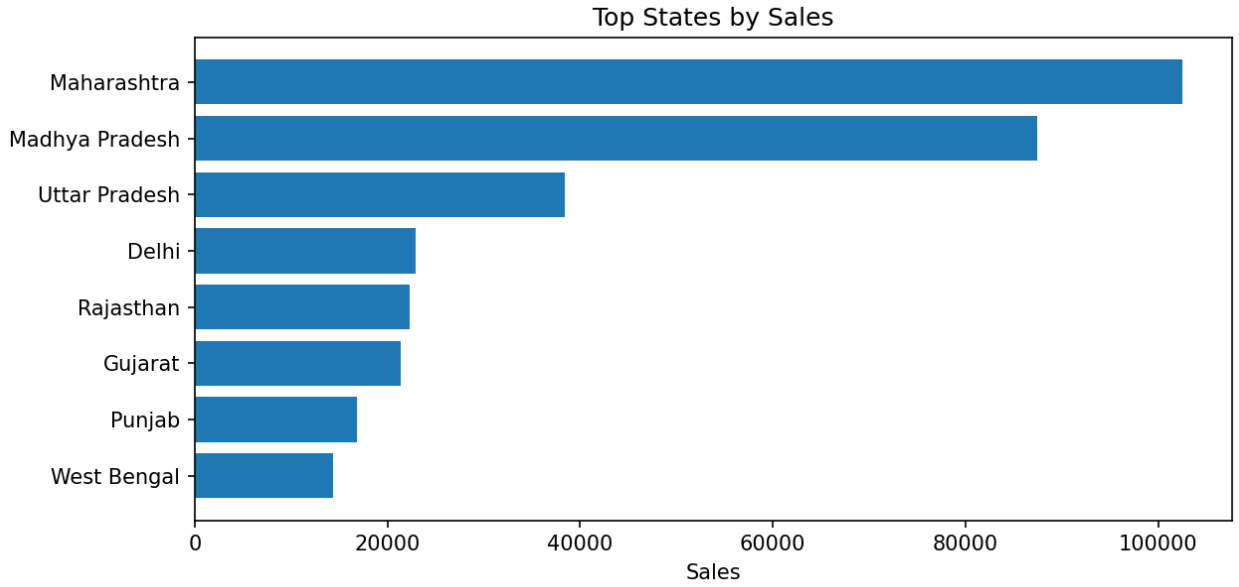
## 3. Key Performance Indicators

| Metric | Value |
|---|---|
| Total Sales | ■437,771 |
| Total Profit | ■36,963 |
| Total Quantity | 5,615 |
| Profit Margin | 8.44% |
| Orders | 500 |
| Detail Records | 1,500 |

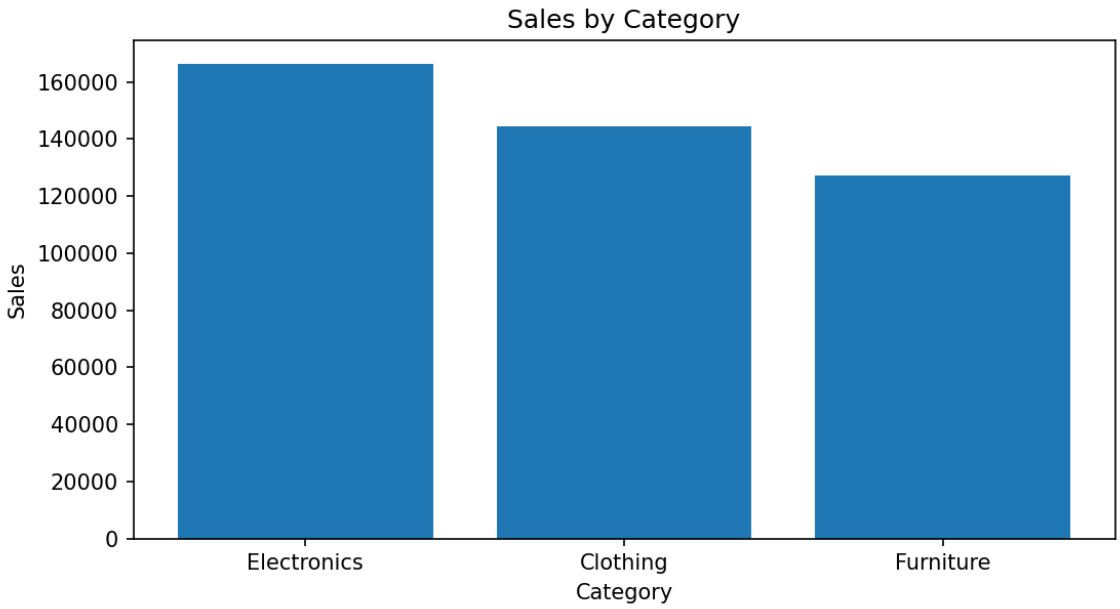
## 4. Exploratory Analysis

The monthly trend varies substantially across the year. January recorded the highest sales in the supplied dataset. Profit should be monitored separately from sales because high sales do not always result in high profit.



## 5. Category Analysis




Clothing is the highest-sales category in the supplied data. The Power BI dashboard provides interactive category and sub-category analysis.

## 6. Machine Learning Component

The Python code includes an exploratory K-Means customer segmentation component. Customers are represented using Recency, Frequency and Monetary value. The features are standardized and grouped into four segments. This demonstrates a basic ML workflow and should be validated with larger, newer data before production use.

## 7. Dashboard

The Power BI dashboard contains KPI cards for Amount, Profit and Quantity plus visuals for state sales, category quantity, monthly profit, customer/state sales, payment modes and category profit.

## 8. Business Insights

- Maharashtra is the highest-sales state.
- Clothing is the highest-sales category.
- January 2018 has the highest monthly sales.
- Profitability should be evaluated separately from sales volume.
- Customer segmentation can support targeted marketing and retention strategies.

## 9. Conclusion

The project demonstrates a complete data analytics workflow: data loading, cleaning, joining, KPI calculation, exploratory analysis, visualization, dashboarding and a basic ML extension.

## 10. Future Enhancements

Add more years of data, validated forecasting, automated refresh and anomaly detection.